

$$V = \underline{11a^2} + \underline{10b^2} + \underline{13c^2} - \underline{4ab} - \underline{6ac} - \underline{20bc} = *$$

$$\begin{aligned} 11a^2 - 4ab - 6ac &= 11\left(a^2 - \frac{4}{11}ab - \frac{6}{11}ac\right) = \\ &= 11\left(\left(a - \frac{2}{11}b - \frac{3}{11}c\right)^2 - \frac{4}{121}b^2 - \frac{9}{121}c^2 - \frac{12}{121}bc\right) \end{aligned}$$

$$* = 11 \cdot \underbrace{\left(a - \frac{2}{11}b - \frac{3}{11}c\right)^2}_{:=A} + \left(10 - \frac{44}{121}\right)b^2 + \left(13 - \frac{99}{121}\right)c^2 + \underbrace{\left(-20 - \frac{132}{121}\right)}_{\cdot b \cdot c}bc$$

$$= 11A^2 + \frac{106}{11}b^2 - \frac{232}{11}b \cdot c + \frac{134}{11}c^2 =$$

$$= 11A^2 + \frac{106}{11} \cdot \left(b^2 - \frac{232}{106}b \cdot c\right) + \frac{134}{11}c^2 =$$

$$= 11A^2 + \frac{106}{11} \cdot \underbrace{\left(b - \frac{116}{106}c\right)^2}_{:=B} - \frac{106}{11} \cdot \frac{58^2}{53^2}c^2 + \frac{134}{11}c^2 =$$

$$= 11A^2 + \frac{106}{11} \cdot B^2 + \frac{34}{53}c^2$$

$$A^2, B^2, c^2 \geq 0$$

$$11, \frac{106}{11}, \frac{34}{53} \geq 0$$

$$\Rightarrow V \geq 0 \quad \forall a, b, c \in \mathbb{R}$$

$$V = 0 \Leftrightarrow a, b, c = 0$$

V je PD